Classifying Sentiment in the Language of PvP Game Communications

Replication and Extension of Borst, Yang, Zhu (Stanford University, 2021)

UE24CS352A Machine Learning Mini-Project | Section C | Team 24 | Project 51
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Code: https://github.com/pranav-codes55/game-chat-toxicity-ml

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1. Problem Statement

In competitive online games like Dota 2 and League of Legends, player chat is fast-paced, highly informal, and frequently toxic. Moderating in-game chat is difficult because players rely heavily on game-specific slang (such as *'ff'*, *'diff'*, *'feeder'*), creative misspellings to bypass profanity filters, and sarcastic flame. Standard sentiment analysis tools trained on formal text fail in this setting. Following the Stanford study by Borst, Yang, and Zhu (2021), this project classifies single in-game chat messages into three categories: **0 - neutral / positive** (tactical calls, praise, e.g., *'gg wp'*, *'push mid'*), **1 - negative attitude** (whining, blaming teammates, giving up, e.g., *'report mid he is feeding'*), and **2 - derogatory** (hate speech, slurs, severe targeted harassment). We compare classical machine learning models on TF-IDF features with neural networks trained on word2vec and GloVe embeddings.

2. Dataset Details

Because the Stanford paper's labelled data is private, we built our own dataset from the two public sources cited in the paper: **GOSU.AI Dota 2 game chats** and **League of Legends Tribunal chat logs** from Kaggle (~1 million messages each). Our data preparation pipeline involved:

- **Cleaning & Filtering:** Repaired encoding issues in the Dota logs, removed non-English and Russian text, removed URLs, and shortened repeated characters (e.g., *'noooob'* → *'noob'*). Punctuation marks (*'?'*, *'!'*) were preserved as sentiment indicators. This yielded **605,771 unique English messages**.
- **Sampling Candidates:** Random chat contains less than 1% derogatory messages. To ensure all three classes appear in sufficient numbers, we sampled **3,400 candidate messages** using a stratified mix: 1,600 random messages, 800 lexicon-matched messages, 600 containing strong insult terms, and 400 from reported tribunal cases.
- **Labelling & Human Audit:** We labelled the first 2,000 candidates using an LLM assistant following written rules. To verify label quality, 97 messages were independently hand-checked blind to the AI labels. The AI agreed with human labels on **76.3% of messages (Cohen's kappa = 0.643)**, showing substantial agreement. We corrected 26 mislabelled cases based on the manual audit.
- **Final Dataset:** After removing 21 invalid fragments, the final dataset has **1,979 messages**: 56.0% neutral/positive (1,108), 36.9% negative attitude (730), and 7.1% derogatory (141). We split this 80/20 into **1,583 training** and **396 test messages** using stratified sampling.

3. Approach

We tested four feature representations across classical and neural models:

- **TF-IDF:** Unigrams and bigrams, sublinear TF scaling, max 3,000 features. Stop words were kept because pronouns (*'you'*, *'i'*) and negations (*'not'*) are essential for identifying personal attacks.
- **165-Phrase Dictionary:** Counts of domain toxic phrases from the Stanford paper's curated lexicon.
- **word2vec:** Skip-Gram embeddings (50 and 100 dimensions) trained on 200,000 unlabelled chat messages from our corpus.
- **GloVe:** Pre-trained 50 and 100-dimensional embeddings from Wikipedia and Gigaword news text.

Embedding	Dimension	Vocab Words Found	Word Occurrences	Nearest Neighbors of 'feeder'
word2vec (our chat data)	50 / 100	99.6 %	99.9 %	*idiot, flamer, noob, troll, retard*
GloVe (Wikipedia / news)	50 / 100	93.7 %	97.8 %	*hogs, cattle, calves, animals*

Table 1. Vocabulary coverage on our test set and nearest neighbours of 'feeder'.

Models evaluated: (1) *Binary SVM baseline* (linear, RBF, polynomial) to test neutral vs. toxic separation; (2) *Classical models on TF-IDF*: Naive Bayes, KNN (K = 1, 2, 3, 6), and Random Forest (Gini, entropy, depth limit 32); (3) *Feedforward neural networks* with 1–2 dense layers (64 and 32 units) over average word vectors; and (4) *LSTMs* with 64 units, 20% dropout, padded to 30 words, with softmax and sigmoid output heads.

4. Brief Overview of Implementation

The project is implemented in Python using pandas, scikit-learn, gensim, and TensorFlow/Keras across seven reproducible notebooks (data preparation, labelling/audit/split, TF-IDF models, embeddings, feedforward NN, LSTM, and evaluation). To prevent data leakage, TF-IDF vectorizers and vocabulary tables were fitted strictly on the training set, and duplicates were removed before splitting. All 12 feedforward and 8 LSTM networks were trained using the Adam optimizer (learning rate 0.001), batch size 64, a 20% validation split for early stopping (patience 8–10, max 100 epochs), and random seed 42. We also built an interactive Streamlit demo (demo/app.py) that allows users to type any chat message and view class predictions and probability breakdowns from both the best TF-IDF model and an LSTM.

Experimental Results

Model	Features	Accuracy	Macro-F1	F1 Derogatory	Paper Acc.	Difference
Binary SVM (linear)*	TF-IDF	0.9217	0.9193	—	0.9565	-0.0348
Naive Bayes (165-phrase dict)	Dictionary	0.8662	0.7075	0.3529	—	—
Naive Bayes (TF-IDF)	TF-IDF	0.8308	0.7513	0.5778	0.9506	-0.1198
KNN (K = 1)	TF-IDF	0.6894	0.5846	0.4186	0.9853	-0.2959
Random Forest (Gini) [Best]	TF-IDF	**0.9015**	**0.8247**	**0.6512**	0.9912	-0.0897
Random Forest (Entropy)	TF-IDF	0.8990	0.8226	0.6512	0.9910	-0.0920
Feedforward NN (ReLU, Sig, Soft)	GloVe 100	0.8182	0.7121	0.4783	0.9572	-0.1390
LSTM (softmax head) [Best NN]	word2vec 100	**0.8434**	**0.7226**	**0.4545**	0.9586	-0.1152
LSTM (softmax head)	GloVe 50	0.8384	0.7123	0.4255	—	—

*Table 2. Test performance on 396 held-out messages (*SVM evaluated on binary neutral vs. toxic task).*

5. Conclusions

- **TF-IDF Models Outperformed Neural Networks:** As in the Stanford paper, Random Forest on TF-IDF achieved the highest accuracy (**0.9015**) and macro-F1 (**0.8247**), beating the best LSTM (0.8434 accuracy, 0.7226 macro-F1) by 5.8% accuracy and 10.2% macro-F1. In fast-paced PvP chat, messages are very short (typically 5 to 8 words). Explicit keywords like *'report'*, *'noob'*, or slurs provide strong linear and tree split signals, allowing decision trees to separate classes effectively without the high parameter count or sample hunger of neural networks.
- **In-Domain word2vec vs. Pre-trained GloVe:** Our chat-trained word2vec embeddings captured game terminology much better (99.9% token coverage) than news-trained GloVe (97.8%). GloVe's nearest neighbours for *'feeder'* were agricultural terms (*'cattle'*, *'hogs'*), whereas word2vec correctly clustered it with *'flamer'* and *'idiot'*. Consequently, models on word2vec achieved higher accuracies than those on GloVe.
- **LSTM vs. Feedforward Networks:** LSTMs outperformed feedforward networks across every embedding setup by 1.8 to 4.0 accuracy points. This confirms that processing word order sequentially is beneficial, whereas feedforward networks lose word order by averaging vectors.
- **Difficulty of the Derogatory Class:** Because derogatory messages make up only 7.1% of the dataset (28 test instances), F1-scores on Class 2 were lower across all models (0.6512 for Random Forest, and below 0.46 for neural networks). Models handled neutral and negative attitude messages reliably (F1 0.87–0.93), but subtle insults and toxic sarcasm without explicit profanity were frequently misclassified.
- **Comparison with Stanford Paper:** Our accuracies are lower than the paper's reported numbers (e.g., 90.15% vs. 99.12% for Random Forest; 68.94% vs. 98.53% for KNN). The paper trained on approximately 33,000 messages (compared to our 1,979) and likely used cleaner or artificially balanced data. Our results reflect realistic performance on unconstrained, noisy game logs.

Team Contributions:

• **R P Pranav (PES2UG24CS917):** Data collection, cleaning and candidate sampling, AI labelling rules, manual audit, label merging, train/test split, and TF-IDF models (SVM, Naive Bayes, KNN, Random Forest).

• **Divyanshi Verma (PES2UG24CS161):** word2vec & GloVe embeddings, feedforward networks, LSTM neural models, evaluation metrics, confusion matrices, comparison with the paper, and Streamlit demo.

Reference & Deliverables:

• **Reference:** Borst, K., Yang, T., Zhu, M. (2021). *Trolls, Haters, or Teammates: Classifying Sentiment in the Language of PvP Game Communications.* Stanford University.

• **Deliverables:** Complete reproducible code, 7 notebooks, serialized models, and Streamlit demo in GitHub repository.